



**PATH TO
THE FUTURE**

Darb Al-Mostaqbal Education Co.
Yenepoya International Schools - KSA



وزارة التعليم
Ministry of Education

STRATEGIC SOLUTION BLUEPRINT

IT OPS OS

Phase 1 - iPad Operations Command Center

Solution Architecture, Infrastructure Design, Resources, Implementation Plan, Governance and Target Outcomes

Document Status	Design Baseline v1.0
Prepared For	Yenepoya International Schools - KSA
Project	IT OPS OS - Phase 1
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Design principle: IT OPS OS is workflow-first, search-first and history-preserving. Inventory supports the workflow; it is not the center of the product.

Document Authority

This document consolidates the current Phase 1 direction approved in this project. Earlier Gemini review PDFs are treated as project-history evidence, not as automatic technical authority. Where earlier archives conflict with the current project direction, this blueprint uses the current decisions agreed in this project.

Executive Summary

Current Situation: Student and faculty iPad operations are distributed across Excel workbooks, paper forms, manual signatures, Microsoft 365, Apple School Manager, Jamf School and human follow-up.

Key Findings: The real operational unit is the workflow: person -> device -> form -> signature -> verification -> lifecycle history. Phase 1 must replace duplicate manual tracking without creating a complicated ERP.

Business Impact: The platform will reduce reconciliation work, improve chain-of-custody visibility, strengthen auditability, reduce security exposure from spreadsheets and provide management with real-time operational status.

Recommendation: Build Phase 1 as one modular cloud application with a shared operational database, Microsoft Entra ID authentication, OneDrive-backed document references, permanent person/device timelines and role-based control.

Expected Benefits: Faster device distribution, fewer missed follow-ups, reliable asset state, automated forms, permanent repair/replacement lineage, controlled access and a platform foundation that can expand to other school IT operations.

Budget Impact: Development can begin using existing Microsoft 365, GitHub and low-cost/free development tiers. Production subscription requirements should be validated after pilot load, security and retention testing.

Timeline: Recommended delivery: design freeze -> foundation -> dashboard/search -> profiles -> assignments/forms -> repair/AppleCare -> return/release/BYOD -> faculty -> integrations -> security/UAT -> pilot/go-live.

Phase 1 succeeds when day-to-day iPad operations can be completed and audited from the portal without depending on Excel as the operational system of record.

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1. Project Purpose and Design Principles

IT OPS OS Phase 1 is a browser-based command center for school iPad operations. Its purpose is to replace fragmented spreadsheet and paper-based tracking with one controlled operational platform while preserving the school's physical-signature requirements.

1.1 Product Definition

A search-first, workflow-driven school IT operations platform that connects people, devices, forms, approvals and permanent lifecycle history into one source of truth.

1.2 Design Principles

- Simplicity first: common actions should require very few clicks and should be understandable without extensive training.
- Workflow first: the system tells users what is pending, what happened and what happens next.
- One authoritative record: a student, faculty member or device is created once; lifecycle events are attached as history.
- No history rewriting: replacements, repairs, returns and corrections create new timeline events rather than overwriting the past.
- Search first: any authorized user should be able to find a person or device quickly from one global query box.
- Security by default: no operational spreadsheet passwords, least privilege, MFA, audit logs and role-aware views.
- Cloud first: no local server dependency and no dedicated on-premises hardware requirement.
- Future-ready: Phase 1 solves iPad operations now while keeping the data and workflow model extensible to future asset categories.

2. Phase 1 Scope and Exclusions

2.1 In Scope

- Dashboard and action queues
- Global search
- Student directory and 360° student profile
- Faculty directory/profile for iPad operations
- iPad registry and 360° device profile
- Student device distribution
- BYOD / migration tracking
- Faculty iPad assignment
- Repair case management
- AppleCare / replacement lineage
- Student/faculty returns
- Permanent release (sale, gift, retirement or ownership transfer)
- Automated operational form generation
- Physical-signature tracking and verification
- OneDrive document references
- Permanent person and device timelines
- Operational reports
- Role-based access, audit trail and security controls

2.2 Explicitly Out of Scope for Phase 1

- Helpdesk and ticketing beyond repair cases
- Full procurement and purchasing workflows
- Finance, accounting, ROI and depreciation

- Payroll and HR system functions
- Facilities operations
- CCTV and access-control operations
- Robotics/STEM/VR/network asset management UI
- Advanced Jamf or Apple School Manager automation
- AI assistant/copilot functions
- Local-server infrastructure
- Password storage or credential history

3. Complete Product Structure

Phase 1 is organized around an experience layer, people, devices, reusable workflows, forms, documents and permanent history.

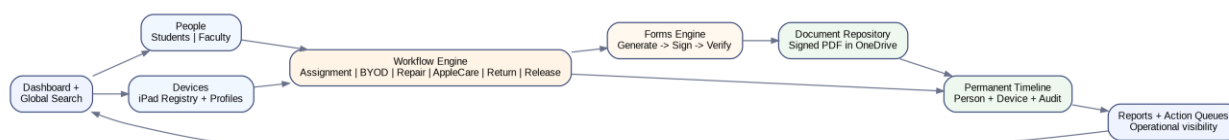


Figure 1 - Phase 1 operating model

3.1 Navigation

Area	Primary Screens	Purpose
Command	Dashboard, Global Search	Understand current status and move directly to work requiring attention.
People	Students, Faculty	360° operational profile, current assignment, documents and timeline.
Devices	iPad Inventory, Device Profile	Current status, owner, AppleCare, history and lineage.
Operations	Distribution, BYOD, Repairs, AppleCare, Returns, Releases	Execute controlled lifecycle workflows.
Documents	Pending Signature, Pending Verification, Archive	Generate forms, track physical signature, verify and retain references.
Reports	Operational Reports	Read-only operational visibility and exports.
Administration	Users, Roles, Configuration, Audit	Govern access and system-level settings.

3.2 Core Screens

Screen	Core Function
Dashboard	Operational KPIs, action required, recent activity and fast links.
Global Search	Search by student/faculty name, ID, email, guardian, serial or asset tag.
Student Profile	Identity, guardian, account status, current device, accessories, workflows, documents and timeline.
Faculty Profile	Employee identity, current/previous device, accessories, documents and timeline.
Device Profile	Permanent device identity, serial, model, coverage, current owner, lifecycle history, documents and lineage.
Workflow Workspaces	Guided steps for assignment, repair, replacement, return and release.



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Screen	Core Function
Forms & Documents	Generated form, signature status, verification metadata and OneDrive reference.
Timeline	Append-only record of material events across a person and device lifecycle.



4. Infrastructure Design

The recommended infrastructure is cloud-native and managed. The school should not operate local application servers, database servers or dedicated hardware for Phase 1.

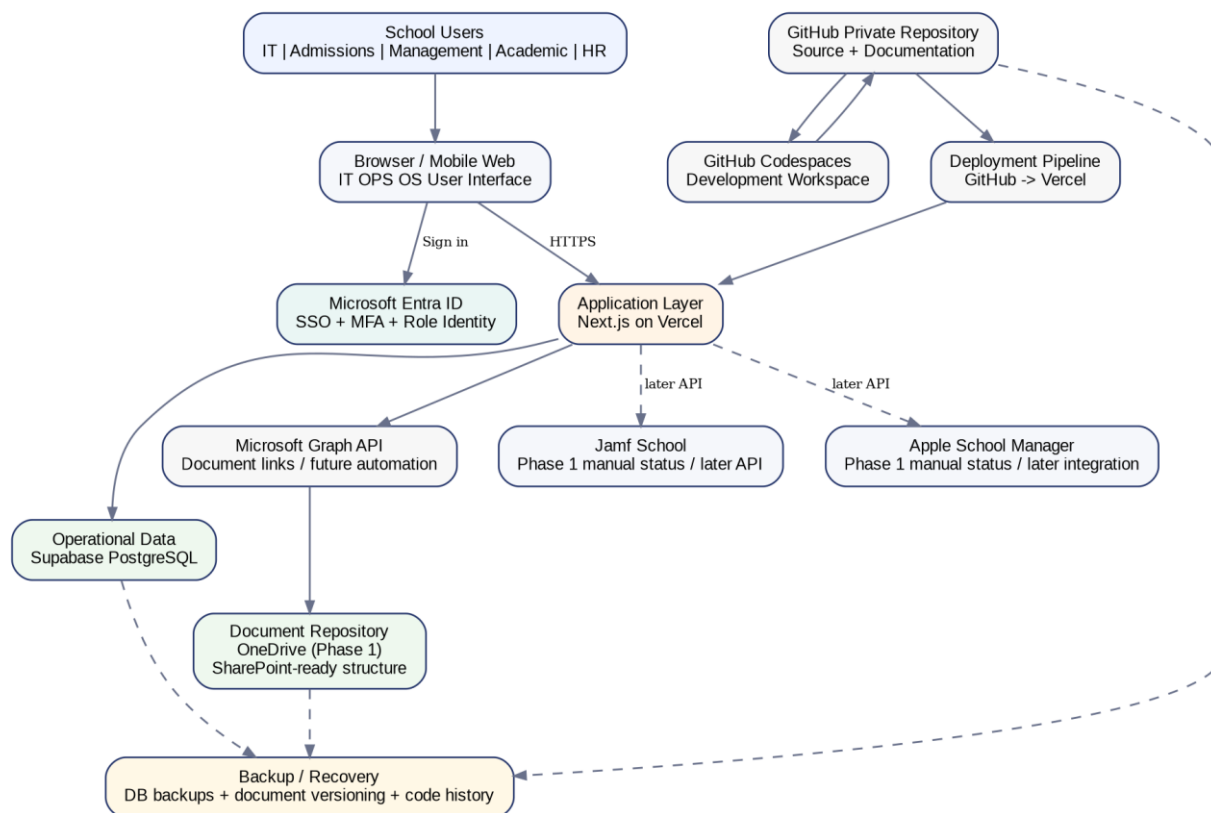


Figure 2 - Phase 1 cloud infrastructure

4.1 Infrastructure Layers

Layer	Phase 1 Design
User Access	Modern browser on Windows, macOS, iPadOS or mobile. No special local client required.
Identity	Microsoft Entra ID provides organizational sign-in, MFA and account lifecycle control.
Application	Next.js web application hosted on Vercel. This is the interface and workflow logic users interact with.
Operational Database	Supabase PostgreSQL stores structured records: people, devices, assignments, workflow state, metadata, timeline and audit.
Documents	OneDrive is the Phase 1 document repository. The portal stores metadata and secure links rather than the signed PDF binary.
Integration	Microsoft Graph for document references and future Microsoft automation. Jamf/ASM integration is staged, not required for first usable release.
Development/Source Control	Private GitHub repository, with GitHub Codespaces available as a cloud development workspace.
Deployment	GitHub-driven deployment to Vercel, with controlled environment variables and release history.

Layer	Phase 1 Design
Recovery	Database backup/restore, OneDrive version history/retention and GitHub source history form the recovery foundation.

4.2 Important Infrastructure Principle

OneDrive is retained as the requested Phase 1 repository. For long-term institutional ownership, the folder structure and application references should be designed so documents can later move to a SharePoint document library without changing business workflows. This avoids tying permanent records permanently to one staff member's personal OneDrive.

5. Resources and Technology Baseline

Resource	What It Is	Phase 1 Role	Status
Next.js	Web application framework	Builds the screens, navigation, forms and server-side application logic.	Selected baseline
Vercel	Managed cloud hosting	Hosts the Next.js application without school-managed servers.	Selected baseline
Supabase PostgreSQL	Managed relational database	Stores structured operational records and relationships.	Selected baseline
Microsoft Entra ID	Identity provider	Single sign-on, MFA and organizational user identity.	Selected baseline
Microsoft 365 / OneDrive	Document repository	Stores approved scanned forms; portal stores links and verification metadata.	Existing school resource
Microsoft Graph	Integration API	Allows controlled Microsoft 365 document/user integrations.	Progressive integration
GitHub Private Repository	Source control and documentation	Stores code, change history, architecture documents and issues.	Selected baseline
GitHub Codespaces	Cloud development workspace	Reduces dependency on one local PC and provides repeatable development environment.	Available resource
Jamf School	MDM	Operational device-management system; status can be recorded manually first and integrated later where supported.	Existing school resource
Apple School Manager	Apple provisioning/ownership	External source for school Apple device lifecycle actions.	Existing school resource

5.1 Cost Position

The development environment can begin with minimal new cost by using existing Microsoft 365, GitHub and free/development tiers where appropriate. Before production go-live, the project should validate expected users, database size, backup retention, document volume, support requirements and vendor production-tier costs. Production should not remain indefinitely on free tiers simply to avoid a small recurring cost if doing so weakens support, security or continuity.

6. Information and Document Architecture

6.1 Core Business Objects

- School / Tenant
- Users and Roles

- Students
- Faculty
- Devices
- Assignments
- Accessories Issued
- Workflows
- Repair Cases
- AppleCare Claims
- Returns
- Releases
- Documents
- Verification Records
- Timeline Events
- Audit Logs

6.2 Data Rules

- One student or faculty member has one master person record.
- One physical iPad has one permanent internal device record.
- Serial number is a device attribute, not the permanent system identity.
- AppleCare replacement creates a new device and links it to the previous device; the old serial is never overwritten.
- Assignment is a relationship/event, not a column that destroys previous ownership history.
- Document state and device state are separate.
- Timeline events are append-only for material history.
- Passwords and authentication secrets are never stored in operational records.

6.3 Document Lifecycle

Generated / Draft -> Pending Physical Signature -> Verified / Approved -> Archived

The actual signed PDF remains in OneDrive. IT OPS OS stores document type, related person/device, generated date, signature-required flag, signature-received date, verification status, verified by, verified date, OneDrive link and archive status.

7. Workflow Architecture

All Phase 1 workflows should use the same reusable operating pattern: collect/confirm information -> perform IT action -> generate required document -> complete physical action -> verify -> update current state -> write permanent history.

Workflow	Controlled Lifecycle
Student Distribution	Student intake -> account/device preparation -> reserve device -> ASM/Jamf/AppleCare checks -> accessories -> generate distribution form -> physical signature -> verify -> assigned.
BYOD / Migration	Register person/device -> confirm ownership/eligibility -> configuration/management actions -> verification -> migration complete.
Faculty Assignment	Faculty intake -> select device -> prepare/configure -> issue accessories -> generate faculty form -> physical approval/signature where required -> verify -> assigned.



Workflow	Controlled Lifecycle
Repair	Issue reported -> create repair case -> condition/coverage check -> repair submission -> device in repair -> outcome recorded -> returned or replacement path.
AppleCare Replacement	Claim -> replacement received -> create new device record -> link old/new lineage -> replacement form -> verify -> new assignment/current device.
Return	Exit trigger -> collect device/accessories -> physical inspection -> Jamf/ASM actions -> return form -> verify -> available/repair/retired outcome.
Permanent Release	Release reason -> verify condition/ownership -> Jamf/ASM release -> release form -> approve -> released/retired history preserved.

8. Security, Roles and Governance

8.1 Security Baseline

- Microsoft Entra ID single sign-on; no separate local staff password database.
- MFA and Conditional Access where configured in Microsoft 365.
- Least-privilege role-based access.
- No plaintext Microsoft 365/email passwords in database, forms, notes, exports or documents.
- HTTPS for all user sessions and integrations.
- Server-side authorization checks for every sensitive create/update/approval action.
- Audit log for critical changes, approvals, assignments, replacements, returns, releases and permission changes.
- Environment secrets stored outside source code.
- Production and test data separated.
- Export permissions restricted; sensitive internal fields excluded from general reports.

8.2 Role Model

Role	Phase 1 Authority
Dashboard Administrator / Head of IT	Full operational and governance control; role assignment; final verification authority.
Authorized IT Team	Execute approved operational workflows according to granted permissions.
Admissions	Read-only student/device operational visibility; optional student intake creation if formally enabled; no device editing.
Management	Broad read-only dashboards, profiles, inventory and authorized reports/exports; no operational editing.
Academic Head	Read-only relevant student/faculty/device visibility; approval only where explicitly assigned.
HR	Faculty intake and relevant faculty-device visibility where needed; no general device administration.

9. Implementation Strategy

Implementation should be delivered in small, testable increments. A working application shell and core data model should be established before attempting external integrations or production data migration.

Stage	Main Work	Exit Result
0. Design Freeze	Approve this blueprint, open decision log, confirm MVP scope, document ownership and acceptance criteria.	Approved baseline; no architecture drift.
1. Foundation	Private GitHub repo, environments, Next.js application shell, Supabase project, Entra authentication skeleton, navigation.	Secure application foundation.
2. Dashboard + Search	Dashboard cards/action queues, global search, synthetic sample data.	Users can navigate and find sample records.
3. People + Device Profiles	Student, faculty and device profile views; status model; timelines.	360° read experience working.
4. Assignment + Forms	Student distribution workflow, accessories, form generation, document status and verification.	End-to-end student assignment works.
5. Repair + AppleCare	Repair cases, coverage checks, replacement lineage, replacement forms.	Full repair/replacement history.
6. Returns + Releases + BYOD	Return inspection, release controls, BYOD/migration workflow.	Core device lifecycle complete.
7. Faculty iPads	Faculty assignment workflow using the same device/document/timeline model.	Student and faculty iPad operations unified.
8. Integrations + Hardening	OneDrive/Graph integration, permission hardening, monitoring, error handling, audit/export controls.	Production-ready controls.
9. UAT + Pilot	User acceptance testing with synthetic/pilot records, training, defects, operating SOPs.	Business approval for controlled go-live.
10. Production Migration + Go-Live	Only after application stability: controlled legacy data migration, reconciliation and production launch.	Portal becomes operational system of record.

9.1 Implementation Rules

- Use synthetic sample data during development until the application is stable.
- Do not make production workbook import the first development task.
- Inspect and preserve the existing repository before replacing or restructuring files.
- Every increment must have a visible expected result and validation checklist.
- One feature is considered complete only after business-rule testing, permission testing and audit verification.
- External integrations must fail safely: users should still understand the workflow state if Jamf/ASM/Graph is temporarily unavailable.

10. Environments, Deployment and Operations

Environment	Purpose
Development	Synthetic data; rapid iteration; developer access only.
Test / UAT	Controlled sample or sanitized pilot records; role testing; management/Admissions/IT validation.
Production	Live school data; restricted administration; production backup/monitoring; formal change control.

Production deployment should be repeatable from the private GitHub repository rather than dependent on one laptop. Configuration secrets must be environment-specific and must never be committed into source control.

11. Testing and Quality Assurance

Test Area	Acceptance Focus
Functional	Each workflow completes correctly, required fields and status transitions are enforced.
Security / RBAC	Users can access only the screens, data and actions authorized for their role.
Data Integrity	Duplicate serials are blocked; old/new device lineage is preserved; current ownership is unambiguous.
Document	Generated forms contain correct stored data; signature/verification state is accurate; OneDrive references work.
Audit	Material changes write complete audit/timeline records with actor and timestamp.
Usability	Common IT tasks can be completed quickly without extensive training.
Browser / Mobile	Supported browsers and iPad/mobile layouts remain usable.
Failure Handling	Unavailable OneDrive/API/integration does not corrupt workflow state or hide errors.
Performance	Search, dashboard and profile pages remain responsive for expected school usage.
UAT	IT, Admissions and management validate their own real-world scenarios before go-live.

12. Backup, Recovery and Business Continuity

- Database backups must be scheduled and restoration tested, not merely assumed.
- OneDrive/SharePoint retention and version history should be configured according to school retention requirements.
- GitHub retains source-code and documentation history; releases should be tagged.
- Recovery documentation must include how to rebuild the production application, restore database state and reconnect document references.
- Critical operational procedures should have a short manual fallback for temporary service outage, followed by reconciliation when the portal returns.
- Target recovery objectives (RPO/RTO) should be approved before production launch.

13. Key Risks and Mitigation

Risk	Rating	Impact	Mitigation
Scope creep	High	Phase 1 becomes a full school ERP before iPad operations work well.	Use formal scope/change control; defer non-iPad modules.
Credential leakage	Critical	Legacy spreadsheet passwords or credentials enter the new system.	Do not migrate password fields; rotate/clean where required.
OneDrive ownership	High	Permanent documents are tied to a personal account.	Use controlled folder governance now; prepare SharePoint migration path.
Architecture drift	High	Framework/database/workflow decisions change during implementation.	Maintain decision log and require owner approval for baseline changes.
Bad legacy data	High	Duplicate/ambiguous serials or historical states pollute production.	Migrate only after application stability; cleanse/reconcile before import.



Risk	Rating	Impact	Mitigation
Over-automation	Medium	Integrations add complexity before core workflows are usable.	Manual status first, API automation only where it reduces proven toil.
Single-person dependency	High	Knowledge remains with one IT person.	GitHub documentation, SOPs, environment/runbook documentation, recoverable cloud infrastructure.
Permission mistakes	High	Read-only roles gain operational edit access.	Server-side RBAC tests and periodic permission review.
Free-tier dependency	Medium	Production exceeds free limits or lacks support.	Validate production tiers before go-live and budget for low-cost managed services.

14. Results Expected at Project Completion

- One browser-based command center for student and faculty iPad operations.
- No routine dependency on Excel for current iPad assignment, repair, replacement, return or release status.
- Global search finds authorized people/devices quickly by name, ID, email or serial number.
- Every student/faculty/device has a 360° operational view with current state and permanent history.
- Device distribution and replacement forms are populated from system data rather than manually retyped.
- Physical signatures remain supported, but signature status and verified scanned documents are traceable digitally.
- AppleCare replacement preserves old and new serial lineage without overwriting history.
- Repair, return and release queues clearly show work pending and overdue.
- Management and Admissions receive controlled read-only visibility without receiving IT editing rights.
- Microsoft 365 passwords are absent from operational data and reports.
- OneDrive document references are attached to related records and audit history.
- An auditable event trail records critical actions, actors and timestamps.
- The application can be redeployed from documented GitHub sources without dependence on one local computer.
- The foundation can accept future asset categories without rebuilding Phase 1 business concepts.

14.1 Management-Level Outcome

Management should be able to ask “What is the status of this student, device, repair, replacement or return?” and receive an immediate answer from one system rather than waiting for IT to reconcile multiple spreadsheets and paper records.

15. Acceptance Criteria / Definition of Done

Area	Done When
Authentication	Authorized staff sign in with Microsoft Entra ID and role restrictions are enforced.
Dashboard	Operational KPIs and action-required queues reflect application data accurately.
Search	Search resolves relevant people and devices using approved identifiers.
Profiles	Student, faculty and device profiles display current and historical operational information.
Assignment	A complete student device assignment can be executed from intake through verified handover.



Area	Done When
Forms	Approved templates are automatically populated and generated without duplicate manual entry.
Documents	Signed form references can be verified and attached to the correct person/device/workflow.
Repair / AppleCare	Repair and replacement cases preserve full device lineage.
Return / Release	Device return and permanent release workflows complete with required checks and resulting device status.
BYOD / Migration	Existing family-owned device onboarding status can be tracked distinctly from school-owned inventory.
Audit / Timeline	Critical events are permanently traceable by actor, date/time and related record.
Security	No plaintext credentials exist in application data, forms or standard exports.
Resilience	Backup and restore process is documented and tested.
UAT	IT and representative read-only stakeholders sign off on agreed scenarios.
Operational Replacement	The portal is accepted as the operational source of truth for Phase 1 iPad workflows.

16. Documentation Completeness Criteria

Yes - your original documentation request was missing several criteria that become important once the project moves from concept to an operational system. They are included in this blueprint so the project has a professional control baseline rather than only a feature list.

Documentation Criterion	Why It Matters
Scope + Exclusions	Prevents Phase 1 from expanding uncontrollably.
Decision / Change Log	Records what was approved, superseded or changed and why.
Assumptions + Constraints	Makes hidden dependencies visible before development.
Non-Functional Requirements	Defines security, performance, availability, usability, privacy and scalability expectations.
Roles / RACI	Clarifies who owns, executes, approves and views each process.
Data Governance	Defines source of truth, ownership, retention, deletion and correction rules.
Security / Privacy	Covers authentication, RBAC, credential handling, audit, secrets and sensitive data.
Integration Map	Shows what is manual now and what may become API-driven later.
Environment Strategy	Separates development, test/UAT and production.
Testing / UAT Plan	Defines how we prove workflows and permissions actually work.
Backup / DR	Defines recovery, retention and outage procedures.
Risk Register	Makes operational and architecture risks trackable.
Cost / TCO	Tracks recurring hosting, support and vendor costs before production commitment.
Deployment / Cutover Plan	Explains how the portal becomes the live operational system without losing records.
Legacy Migration Plan	Deferred until stable, but must exist before Excel is retired.



Documentation Criterion	Why It Matters
Training + SOPs	Ensures staff can use workflows consistently.
Operations / Maintenance Runbook	Documents monitoring, updates, incidents, access review and owner handover.
Definition of Done	Prevents “dashboard loads” from being mistaken for project completion.
Post-Go-Live Review	Measures adoption, defects, manual work removed and Phase 2 readiness.

16.1 Recommended Project Document Set

- 01 - Project Control Baseline
- 02 - Phase 1 Solution Architecture & Implementation Blueprint (this document)
- 03 - Functional Requirements Specification
- 04 - UI/UX Screen Specification
- 05 - Workflow & Business Rules Catalogue
- 06 - Data Dictionary / Database Design
- 07 - Forms Template Register
- 08 - Roles & Permissions Matrix
- 09 - Security & Privacy Design
- 10 - Test & UAT Plan
- 11 - Deployment / Cutover Plan
- 12 - Backup & Disaster Recovery Runbook
- 13 - Operations & Maintenance SOP
- 14 - Legacy Migration & Reconciliation Plan
- 15 - User Guide / Training Pack
- 16 - Decision Log + Risk Register + Change Log

17. Glossary for Non-Developers

Term	Simple Meaning
Frontend	The screens and buttons users see in their browser.
Backend	The hidden application logic that checks rules and performs actions.
Database	The organized storage for structured operational information.
API	A controlled way for one software system to talk to another software system.
Authentication	Proving who the user is, for example through Microsoft Entra ID.
Authorization / RBAC	Deciding what an authenticated user is allowed to see or change based on role.
Hosting	The cloud service where the application runs and becomes accessible on the internet.
Deployment	Publishing a tested application version into an environment such as production.
Repository	The controlled GitHub location containing source code and project documentation.
Webhook	A message one system can automatically send another system when an event happens.
Audit Log	A protected record of who changed what and when.
RPO / RTO	Recovery targets: how much recent data loss is acceptable and how quickly service must be restored.

Final Recommendation

Freeze this blueprint as the Phase 1 solution baseline before coding. The next design artifact should define the exact Application Shell, Navigation and Dashboard specification, followed by the underlying data model and workflow/business-rule catalogue. Implementation should begin only after those baselines are approved enough to prevent another architecture drift cycle.

Build the smallest complete operational loop first: Login -> Dashboard/Search -> Student -> Device -> Assignment -> Form -> Physical Signature -> Verification -> Timeline. Then add repair, AppleCare, return, release, BYOD and faculty workflows on the same foundation.